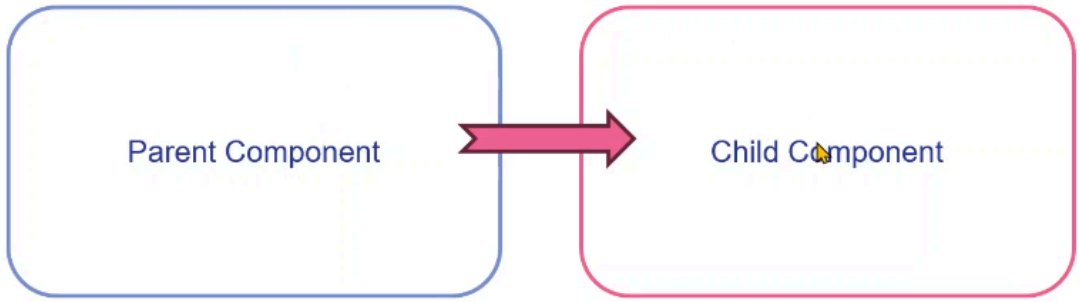
Props:-
Props (short for “properties”) are how you pass data from one component to another in React.

In simple terms its like “**A parcel of data sent from a parent component to a child component.**”



Example:-

Parent Component:-
```
function App() {
  return (
    <User name="Vamshi" age={20} />
  );
}
```

Child Component:-
```
function User(props) {
  return (
    <h1>
      Name: {props.name}, Age: {props.age}
    </h1>
  );
}
```

Instead of writing props.name, we can also write it as:-
```
function User({ name, age }) {
  return (
    <h1>
      Name: {name}, Age: {age}
    </h1>
  );
}
```

Note:-

- Props are read-only
- Child components cannot modify props
- Props make components reusable
- Data flows one way (Parent → Child)
- Without props every component would show static data.With props same component can show different data
- Like same component different outputs.

```
<User name="Rahul" age={22} />
<User name="Anita" age={19} />
<User name="Kiran" age={21} />
```

Exam.jsx

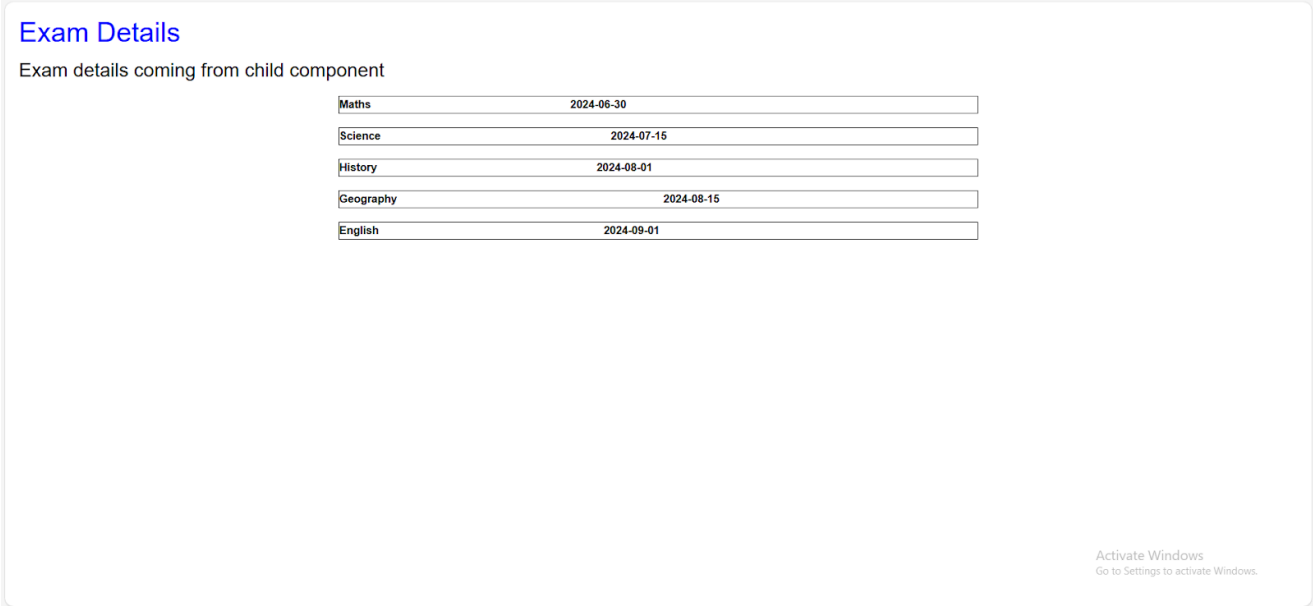
```
import React from 'react'

function Exam(props) {
  return (
    <>
    <table style={{ border: '1px solid black', width: '50%', margin: '20px
auto' }}>
      <tr>
        <th>{props.subject} </th>
        <th>{props.date}</th>
      </tr>

    </table>
    </>
  )
}

export default Exam
```

Student.jsx

```jsx
import React from 'react'
import Exam from './Exam.jsx'
function Student() {
  return (
    <>
      <h1 style={{ color: 'blue' }}>Exam Details</h1>
      <h3>Exam details coming from child component</h3>
      <Exam subject="Maths" date="2024-06-30" />
      <Exam subject="Science" date="2024-07-15" />
      <Exam subject="History" date="2024-08-01" />
      <Exam subject="Geography" date="2024-08-15" />
      <Exam subject="English" date="2024-09-01" />
    </>
  )
}

export default Student
```

Main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import Student from './components/Student.jsx'

// import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <Student />
  </StrictMode>,
)
```



React Jumbotron:-

In Bootstrap, a Jumbotron is a large hero section used to highlight important content — usually placed at the top of a webpage.

It typically contains:

- A big heading
- A short description
- A call-to-action button

A reusable React component that displays a large hero-style section (inspired by Bootstrap's Jumbotron) to highlight important content.

Example:-

```
function Jumbotron() {
  return (
    <div style={{
      padding: "50px",
      backgroundColor: "#f8f9fa",
```

```
      textAlign: "center",
      borderRadius: "10px"
    }}>
      <h1>Welcome to My Website</h1>
      <p>This is a simple hero section built in React.</p>
      <button style={{ padding: "10px 20px" }}>
        Learn More
      </button>
    </div>
  );
}

export default Jumbotron;
```

Then we can use it inside main.jsx:-

```
import Jumbotron from "./Jumbotron";
function App() {
  return (
    <div>
      <Jumbotron />
    </div>
  );
}
```

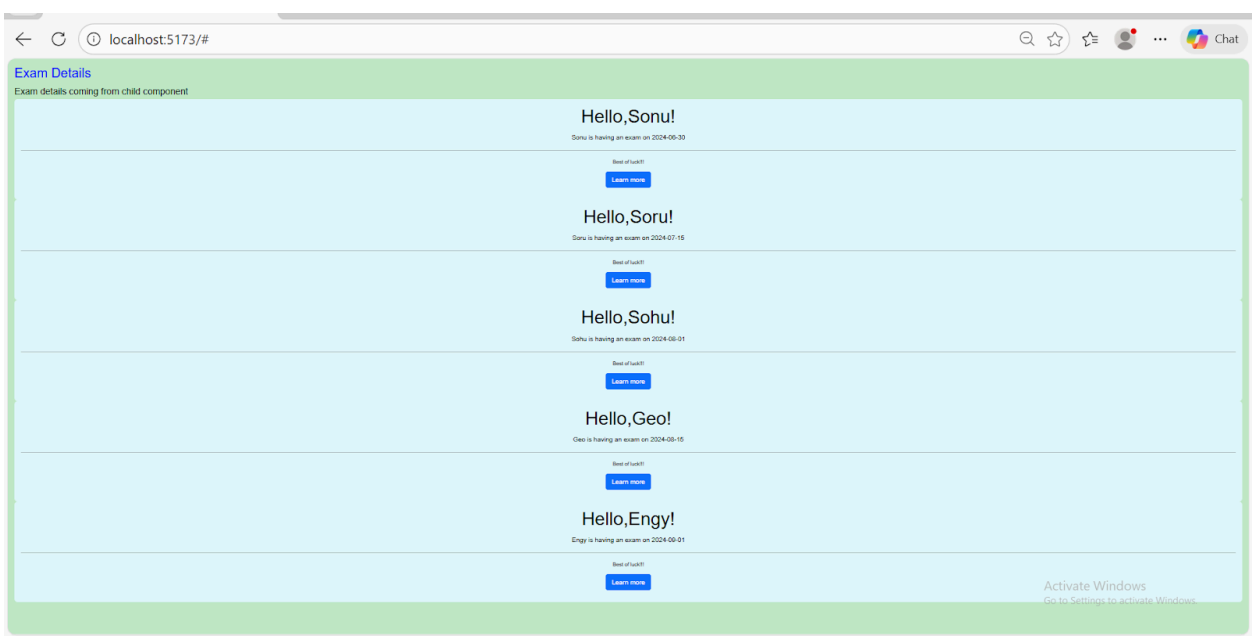
Exam.jsx

```
import React from 'react'


function Exam(props) {

  return (


    <div className="jumbotron" style={{ backgroundColor: '#e0f7fa',
padding: '20px', borderRadius: '10px', textAlign: 'center' }}>

  <h1 className="display-4">Hello,{props.name}!</h1>

  <p className="lead">{props.name} is having an exam on {props.date}</p>

  <hr className="my-4"></hr>

  <p>Best of luck!!!</p>

  <p className="lead">

    <a className="btn btn-primary btn-lg" href="#" role="button">Learn
more</a>

  </p>

</div>

  )

}


export default Exam
```

Student.jsx

```
import React from 'react'

import Exam from './Exam.jsx'

function Student() {
```

```
  return (
    <>
      <h1 style={{ color: 'blue' }}>Exam Details</h1>
      <h3>Exam details coming from child component</h3>
      <Exam name="Sonu" date="2024-06-30" />
      <Exam name="Soru" date="2024-07-15" />
      <Exam name="Sohu" date="2024-08-01" />
      <Exam name="Geo" date="2024-08-15" />
      <Exam name="Engy" date="2024-09-01" />
    </>
  )
}

export default Student
```



useRef Hook:-

useRef is a React Hook that:

1. Stores a value without re-rendering the component
2. Directly accesses a DOM element

Syntax:-

import { useRef } from "react";

const myRef = useRef(initialValue);

We can access value using :-

myRef.current

2 main uses of useRef are: -

->Accessing DOM Elements

->Store value without re-render

When to use useRef:-

1. You need DOM access (focus, scroll, video play)

2.You want to store previous value

3. You want to store timer ID

4. You want to avoid unnecessary re-renders

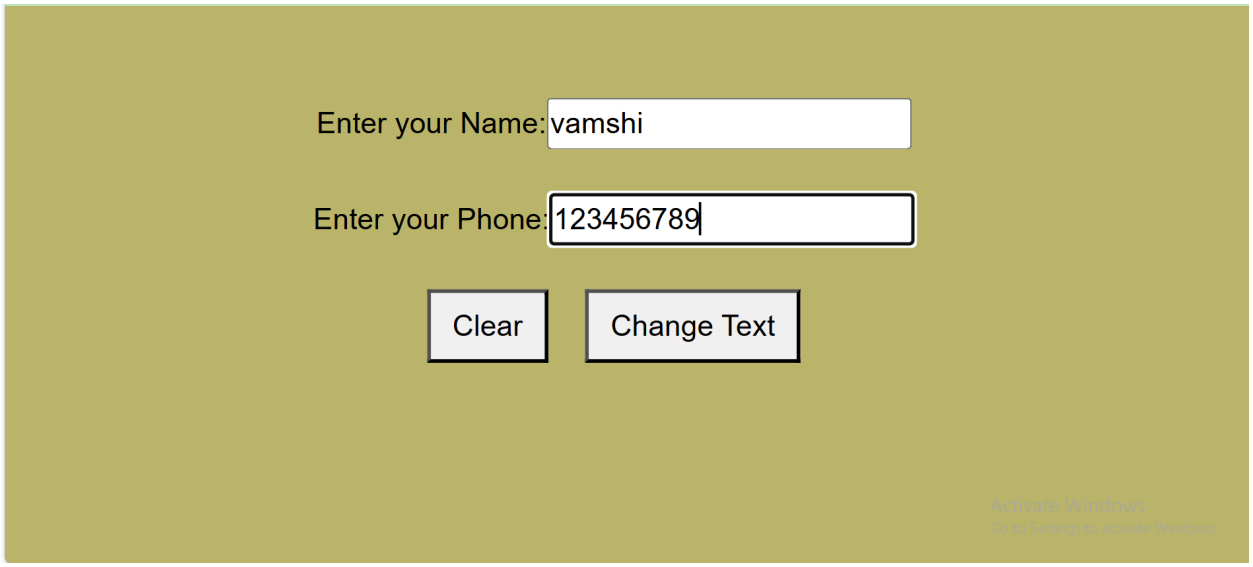
Its an alternative to **document.getElementById()**

Form.jsx

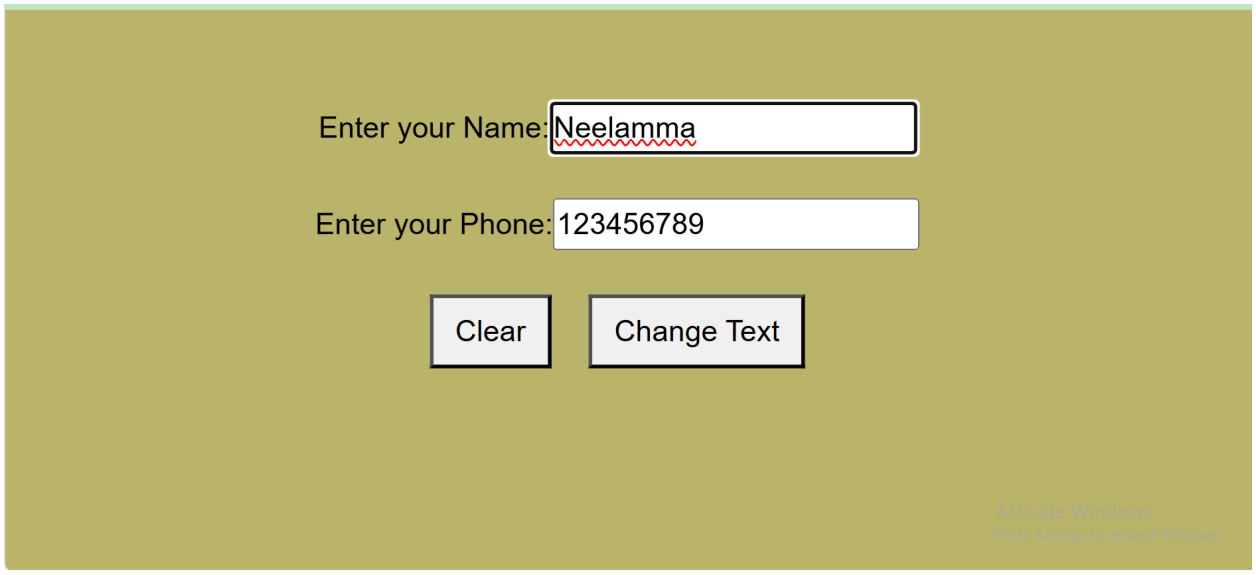
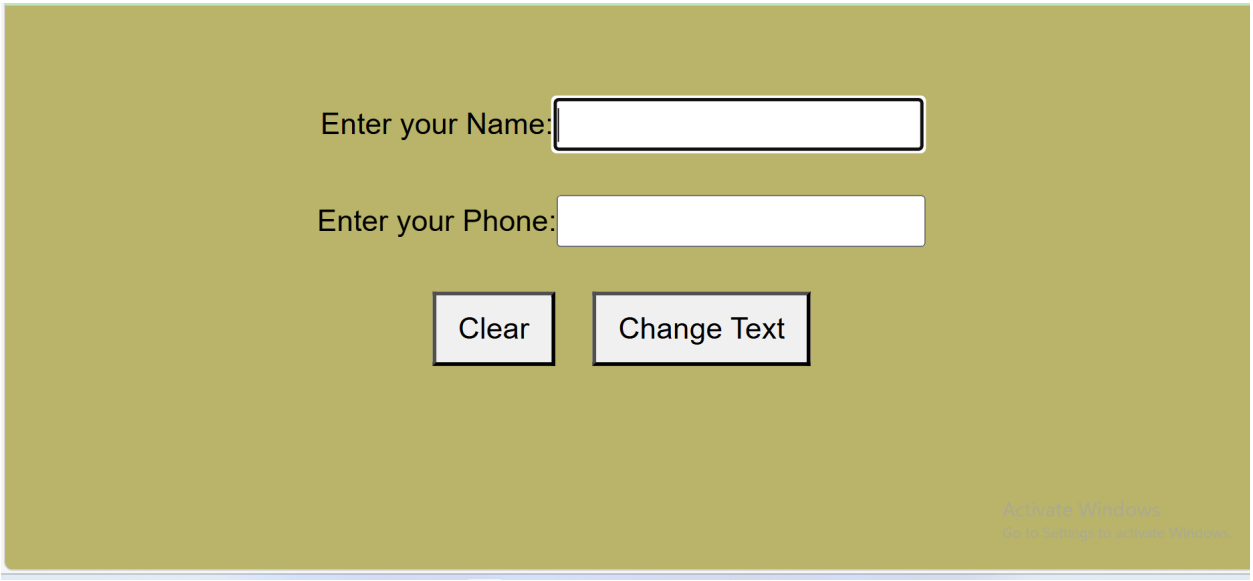
```
import React, { useRef } from "react";

function StudentForm() {
```

```
const nameRef = useRef(null);
const phoneRef = useRef(null);

const handleClear = () => {
  nameRef.current.value = "";
  phoneRef.current.value = "";
  nameRef.current.focus(); // cursor back to name
};

const handleChangeText = () => {
  nameRef.current.value = "Neelamma";
  nameRef.current.focus(); // keep cursor in name field
};

return (
  <div
    style={{
      backgroundColor: "#bdb76b",
      height: "100vh",
      paddingTop: "50px",
      textAlign: "center"
    }}
  >
    <div>
      <label>Enter your Name: </label>
      <input type="text" ref={nameRef} />
    </div>

    <br />

    <div>
      <label>Enter your Phone: </label>
      <input type="text" ref={phoneRef} />
    </div>

    <br />

    <button
      onClick={handleClear}
```

```
        style={{ marginRight: "20px" }}
      >
        Clear
      </button>

      <button onClick={handleChangeText}>
        Change Text
      </button>
    </div>
  );
}

export default StudentForm;
```



When clicked change text:-



When clicked clear:-